Capital Budgeting & Asset Accounting

Finance certification — quantitative section (sample) · Sample Examining Board

EXAM DATE	DAYS REMAINING	TARGET SCORE
2026-12-05	D-119	85
PASSING CRITERION	**SYLLABUS / VERSION**	**CURRENT CONFIDENCE**
70% overall, no section below 50%	2026 curriculum, Reading 21–24 (sample — verify)	●●●○○ 3/5

What does this exam actually test?

- Whether you pick the right method under a stated constraint, not whether you can recite it
- Whether you notice the assumption that makes a method invalid
- Unit and sign discipline under time pressure
- Whether you can rank projects when two methods disagree

SUBJECTS

- Corporate finance
- Financial reporting
- Quantitative methods

SOURCE MATERIALS

- Official curriculum readings
- Past papers 2021–2025

The cycle

LEARN › STRUCTURE › COMPARE › RETRIEVE › APPLY › CORRECT › COMPRESS

> CORRECT is the load-bearing step. A wrong answer edits a named matrix cell, and that cell comes back in the next recall pass.

How to read a cell

Markers are encoded three ways — label, border, tint — so nothing is lost in grayscale.

Marker	Means	What to do with it
CORE	Baseline fact	Know it. Low retrieval priority.
◆ DIST	This is what separates the rows	Be able to state it in one clause.
▲ EXC	The rule does not hold here	Memorise the boundary, not the rule.
■ TRAP	Examiners build wrong answers from this	Rehearse the wrong version and why it is wrong.
↻ UPD	Version-sensitive — verify the date	Re-check against the current text before the exam.
§ EVD	Wording or source, quoted	Recognise the phrasing on sight.

EXAM MATRIX

Exam Blueprint

Capital Budgeting & Asset Accounting

Domain	Weight	Question type	Priority	Current confidence	Last review
Capital budgeting	35%	Item set, 6 vignettes	A	●●●○○ 3/5	2026-07-29
Cost of capital	20%	Multiple choice	B	●●●●○ 4/5	2026-07-15
Long-lived assets	25%	Item set, 4 vignettes	A	●●○○○ 2/5	2026-08-04
Working capital	20%	Multiple choice	C	●●●●○ 4/5	2026-06-30

HIGH-YIELD TOPICS

- NPV vs IRR conflict on mutually exclusive projects
- Depreciation effect on early-year net income

WEAK TOPICS

- Multiple IRR under non-conventional cash flows
- Units-of-production with revised estimates

VOLATILE / VERSION-SENSITIVE

- Any change to the standard-setter guidance on componentisation

Topic index

#	Topic	Archetype	Priority	Rows
01	**Capital budgeting decision rules** What each method assumes about reinvestment and scale	Calculation / Engineering / Finance	A	4
02	**Depreciation methods** Front-loaded vs straight-line expense recognition	Calculation / Engineering / Finance	A	4

TOPIC 01 · CALCULATION / ENGINEERING / FINANCE

A

Capital budgeting decision rules

Every paper, 2021–2025
Last updated: 2026-07-29

Key comparison axis: What each method assumes about reinvestment and scale

WHY IT MATTERS

Four rules that usually agree and occasionally do not. The exam lives entirely in the cases where they disagree, and asks which one you trust.

CORE QUESTION

When these methods rank two projects differently, which one is right and why?

PREREQUISITES Time value of money · WACC **SOURCE** Curriculum reading 21

One-sentence model

Explain this topic in ONE sentence, as if to someone sitting the exam tomorrow. If you cannot, you do not know it yet.

NPV measures value added in currency and is always correct for mutually exclusive projects; the others are shortcuts that fail in a specific, predictable way.

Five essential facts

Maximum five. If you want a sixth, one of these was not essential.

1. NPV assumes reinvestment at the cost of capital; IRR assumes reinvestment at the IRR.
2. For mutually exclusive projects, always follow NPV when the two conflict.
3. Non-conventional cash flows can produce multiple IRRs, or none.
4. Payback ignores everything after the cutoff and ignores time value entirely.
5. PI and NPV always agree on accept/reject; they can disagree on ranking.

Capital budgeting decision rules

Capital budgeting decision rules — Exam Matrix

Method	What it gives you	Decision rule	◆ DIST Key assumption	◆ DIST When to use it	▲ EXC When it breaks	■ TRAP Common mistake
NPV Net present value	Value added, in currency	Accept if NPV > 0	Reinvestment at the cost of capital	Default. Always correct for mutually exclusive choices.	Under hard capital rationing, ranking by NPV alone can waste budget	Discounting at the IRR instead of the WACC
IRR Internal rate of return	A percentage return	Accept if IRR > required return	Reinvestment at the IRR itself — usually unrealistic	Independent projects with conventional cash flows	Non-conventional cash flows → multiple or no IRR	Using IRR to rank mutually exclusive projects of different scale
Payback Payback period	Time to recover the outlay	Accept if below the chosen cutoff	Nothing about time value at all	Rough liquidity screen only	Ignores every cash flow after the cutoff	Treating a short payback as evidence of profitability
PI Profitability index	PV of inflows per unit of outlay	Accept if PI > 1	Same as NPV	Capital rationing — ranking by value per unit of scarce capital	Can rank against NPV when project scales differ	Forgetting that PI > 1 and NPV > 0 are the same condition

CORE core DIST ◆ distinction EXC ▲ exception TRAP ■ trap UPD ↻ update EVD § evidence

DETAIL I ADDED MYSELF

Capital budgeting decision rules

Decisive distinction

If an examiner forced me to separate these using ONE criterion, what would it be?

What the method assumes you do with the money that comes back. NPV assumes you earn the cost of capital on it; IRR assumes you earn the IRR. Every ranking conflict between them traces to that one assumption.

Confusion pairs

A	B	Why confused	Decisive clue
NPV	**IRR**	They agree on accept/reject for a single conventional project, so students assume they always agree.	If the projects are mutually exclusive or differ in scale or timing, follow NPV.
PI	**NPV**	Both are NPV-based, so they are assumed interchangeable.	PI is a ratio. Ratios rank by efficiency, not by size — that is the whole difference.

Exception & trap matrix

Only facts that can change an answer choice.

Statement / cue	Usually true	Exception	Why it tricks people	Source
■ The project with the higher IRR should be selected.	True for independent projects	False for mutually exclusive projects — select on NPV	The percentage feels more comparable than a currency amount.	Reading 21
■ A project with a positive NPV always has an IRR above the cost of capital.	True	Not defined when cash flows are non-conventional and no real IRR exists	Students generalise from the conventional single-sign-change case.	Reading 21
■ Discounted payback fixes the flaws of payback.	Partly true	It corrects for time value but still ignores all cash flows after the cutoff	"Discounted" is read as "now it is a proper NPV method".	Reading 21

Question & evidence links

Matrix cell	Source	Exam / year	Question	Tested as	Error risk
IRR · Common mistake	Reading 21	Sample 2024	Item set 3, Q2	`"...the analyst should select Project B because its IRR is higher"`	High
Payback · When it breaks	Reading 21	Sample 2023	Q17	`"...does not consider cash flows beyond the payback period"`	Mid

TOPIC 02 · CALCULATION / ENGINEERING / FINANCE

A

Depreciation methods

4 of last 5 papers

Last updated: 2026-08-04

Key comparison axis: Front-loaded vs straight-line expense recognition

WHY IT MATTERS

The arithmetic is easy; the exam tests which line of the financial statements moves, in which direction, in which year.

CORE QUESTION

In the first year, which method reports the lowest net income — and why does that reverse later?

PREREQUISITES Carrying amount · Salvage value **SOURCE** Curriculum reading 24

One-sentence model

Explain this topic in ONE sentence, as if to someone sitting the exam tomorrow. If you cannot, you do not know it yet.

Accelerated methods move expense into early years, so early net income and early carrying amount are lower, and both reverse before the asset's life ends.

Five essential facts

Maximum five. If you want a sixth, one of these was not essential.

1. Total depreciation over the asset's life is identical under every method.
2. DDB ignores salvage value in the rate but cannot depreciate below it.
3. Straight line gives the lowest first-year expense of the three.
4. Units-of-production tracks usage, so it is the only method insensitive to time.
5. A change of estimate is applied prospectively, never by restating prior years.

Depreciation methods

Depreciation methods — Exam Matrix

Method	Formula	TRAP Salvage treatment	DIST Early-year expense	TRAP Common mistake
Straight line	(Cost − Salvage) ÷ Useful life	Subtracted up front	Lowest	Using it as the default when the vignette states usage-based consumption
Double declining balance DDB	2 ÷ Useful life × Beginning carrying amount	Excluded from the rate, but is a floor	Highest	Subtracting salvage before applying the rate
Sum-of-years digits SYD	(Cost − Salvage) × Remaining life ÷ Sum of years	Subtracted up front	High, but less front-loaded than DDB	Using remaining life for the numerator of the wrong year
Units of production	(Cost − Salvage) × Units this period ÷ Total estimated units	Subtracted up front	▲ Depends on usage, not on time	Failing to revise the rate when total estimated units change

CORE core   DIST ◆ distinction   EXC ▲ exception   TRAP ■ trap   UPD ↻ update   EVD § evidence

DETAIL I ADDED MYSELF

Depreciation methods

Decisive distinction

If an examiner forced me to separate these using ONE criterion, what would it be?

Whether salvage value enters the calculation at the start. Every method subtracts it up front except DDB, which uses it only as a floor.

Confusion pairs

A	B	Why confused	Decisive clue
DDB	**SYD**	Both are accelerated, so they are treated as interchangeable.	If salvage is missing from the first-year computation, it is DDB.

Exception & trap matrix

Only facts that can change an answer choice.

Statement / cue	Usually true	Exception	Why it tricks people	Source
■ Accelerated depreciation reduces total reported income.	False	It only shifts income between periods — the lifetime total is unchanged	Lower early income is mistaken for lower total income.	Reading 24
■ A change in useful life requires restating prior periods.	False	It is a change in estimate, applied prospectively	Confused with a change in accounting policy, which can require restatement.	Reading 24

Question & evidence links

Matrix cell	Source	Exam / year	Question	Tested as	Error risk
Double declining balance · Salvage treatment	Reading 24	Sample 2025	Item set 2, Q4	`"...applies the rate to the carrying amount at the beginning of the period"`	**High**

EXAM MATRIX

Error log

Every conceptual mistake must name the matrix cell it invalidates.

Date	Question	Topic	My answer	Correct	Error type	Missing distinction	Matrix update	Retest
2026-07-31	Sample 2024, Item set 3 Q2	Capital budgeting	Project B (higher IRR)	Project A (higher NPV)	`Confused pair`	Mutually exclusive → NPV governs, regardless of IRR	Capital budgeting decision rules · IRR · Common mistake	2026-08-14
2026-08-04	Sample 2025, Item set 2 Q4	Depreciation	(Cost − Salvage) × 2/n	Carrying amount × 2/n	`Formula selection`	DDB excludes salvage from the rate	Depreciation methods · Double declining balance · Salvage treatment	2026-08-18

· Knowledge gap · Confused pair · Exception forgotten · Wording misread · Formula selection · Calculation error · Procedure order · Outdated information · Careless · Time pressure

EXAM MATRIX

Compression ladder

Capital Budgeting & Asset Accounting

L1	L2	L3
Full Matrix Every distinction, every exception, every source.	**Exam-day Matrix** High-yield distinctions, exceptions, formulas, traps.	**Last 10 minutes** One page. What you would forget under pressure.

Exam-day matrix — L2

Markers	Topic	Decisive clue
◆ DIST	**Mutually exclusive**	Follow NPV. Always.
◆ DIST	**Reinvestment assumption**	NPV → cost of capital. IRR → the IRR itself.
▲ EXC	**Multiple IRR**	Only with non-conventional cash flows (more than one sign change).
■ TRAP	**DDB and salvage**	Not in the rate. Is a floor.
■ TRAP	**Change of estimate**	Prospective. Never restate.
CORE	**Lifetime depreciation**	Identical under every method.

Last-10-minutes sheet — L3

- [ ] Conflict → NPV wins
- [ ] IRR assumes reinvestment at IRR
- [ ] Sign changes > 1 → multiple IRR
- [ ] PI is a ratio, ranks by efficiency
- [ ] DDB - carrying amount x 2/n
- [ ] Salvage is a floor, not a subtraction (DDB)
- [ ] Estimate change - prospective only
- [ ] Total depreciation is method-independent

Review tracker

Dates or ticks. Keep it secondary to the content.

Topic	Learned	R1	R2	R3	Mastered	Revisit
01 Capital budgeting decision rules						
02 Depreciation methods						